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Human Kinetics
SYSTEMATIC REVIEW

# A Systematic Review of Rehabilitation Exercises to Progressively Load the Gluteus Medius

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***Context:*** Gluteus medius rehabilitation is of critical importance given its role in pelvic and lower limb stability, and the known link between gluteus medius weakness and many lower limb conditions. ***Objective:*** To systematically review the literature and present an evidence-based graduated series of exercises to progressively load gluteus medius. ***Evidence Acquisition:*** A systematic literature search was conducted in January 2016 to identify studies reporting gluteus medius muscle activity as a percentage of maximal volitional isometric contraction (MVIC), during rehabilitation exercises. Studies that investigated injury free participants were included. No restrictions were placed on the type or mode of exercise, though exercises that could not be accurately replicated or performed within an independent setting were excluded. Studies that did not normalize electromyographic activity to a side lying MVIC were excluded. Exercises were stratified based on exercise type and %MVIC: low (0% to 20%), moderate (21% to 40%), high (41% to 60%), and very high (> 61%). ***Evidence Synthesis:*** 20 studies were included in this review, reporting outcomes in 33 exercises (and a range of variations of the same exercise). Prone, quadruped, and bilateral bridge exercises generally produced low or moderate load. Specific hip abduction/rotation exercises were reported as moderate, high, or very high load. Unilateral stance exercises in the presence of contralateral limb movement were often high or very high load activities, while high variability existed across a range of functional weight-bearing exercises. ***Conclusions:*** This review outlined a series of exercises commonly employed in a rehabilitation setting, stratified based on exercise type and the magnitude of gluteus medius muscular activation. This will assist clinicians in tailoring gluteus medius loading regimens to patients, from the early postoperative through to later stages of rehabilitation.

***Keywords:*** electromyography, hip abductors

Therapeutic rehabilitation exercises of the hip abductors, with particular reference to gluteus medius, are commonly prescribed by therapists to improve strength and facilitate more favorable lower extremity movement patterns. This is of particular importance given their role in maintaining a level pelvis and preventing hip adduction and femoral internal rotation during single limb support,[1,2] as well as the known link between gluteus medius weakness and a range of pathologies including lateral hip pain,[3] knee osteoarthritis,[4] patellofemoral pain,[5–7] and chronic low back pain.[8] Many studies appear to focus on investigating exercises that may permit gluteus medius strength gains and make recommendations as to the best exercises to maximally load gluteus medius.[9–13] However, when we consider that hip abductor weakness[14] and gluteus medius atrophy[15] is observed in patients with gluteal tendon pathology, it is plausible that a more graduated loading protocol that considers the full spectrum of gluteal loading exercises is required in patients with gluteal pathology and/or following surgical repair.

Electromyography (EMG) provides a measure of muscle activation and it is assumed that activities eliciting higher EMG signal amplitudes create the potential for greater strengthening effects.[9] Certainly, we may seek to intervene with exercises that maximally load gluteus medius at later stages of rehabilitation, given it has been suggested that muscular activation levels ≥ 40% maximal volitional isometric contraction (MVIC) are required for strength gains.[9,16,17] However, while these exercises may be appropriate to maximally load gluteus medius at later stages of rehabilitation, they are likely inappropriate for patients who present with significant hip abductor atrophy[15] and/or weakness[14] as a result of a chronic injury, or are early postsurgery. We may instead be attempting to improve muscular endurance, prevent physical de-conditioning and/or facilitate motor control or neuromuscular activation,[9,18] using low level exercises in the earlier rehabilitation stages. It has been previously reported that muscle activity < 25% MVIC functions in an endurance capacity or to maintain stability.[17] Therefore, previous studies have stratified exercises into low (0% to 20% MVIC), moderate (21% to 40% MVIC), high (41% to 60% MVIC), and very high (> 61% MVIC) loading

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groups,[19,20] providing a means to guide the loading progression of a clinical rehabilitation program.

# Objective

The purpose of this manuscript was to systematically review the current literature investigating gluteus medius muscular activation (peak EMG activity as a percentage of MVIC) during a range of commonly prescribed rehabilitation exercises. This will provide the clinician with the evidence-based ability to better develop a graduated approach to loading gluteus medius, from the early postoperative through to later stages of rehabilitation.

# Evidence Acquisition

## Database Sources and Search Criteria

A systematic review was undertaken following the guidelines outlined in the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analyses) statement[21] where possible. A comprehensive search of electronic databases (Medline, SPORTDiscus, Scopus, CINAHL, EMBASE) was undertaken to obtain all literature up until January 2016. Each database was searched using the following combined and/or truncated search terms: (glut* OR medius OR hip abduct*) AND (exercis* OR resist* OR rehab* OR physical therapy OR strength* OR load* OR training) AND (electromyo* OR EMG). The asterisk denotes a truncated search term and subsequent inclusion of all related terms that begin with that term. Articles from search databases were exported into a designated EndNote X7 (Endnote, Thomson Reuters, New York, NY) library, so that duplicate references could be removed, and study titles and abstracts screened.

## Inclusion and Exclusion Criteria

A specific quality assessment tool could not be appropriately applied to this review, nor were studies assessed for risk of bias, given this was not a systematic literature review of randomized controlled trials. However, a specific set of study inclusion/exclusion criteria were developed. Articles written in English and recruiting healthy, injury free participants of any age or gender were included, as were studies that evaluated gluteus medius activity via surface or fine-wire EMG methods. There was no restriction on the type of exercise evaluated and, while exercises requiring the addition of external load (ie, cuff weights, theraband) were included provided the load could be replicated, exercises that could not be accurately replicated or could not be performed within an independent home-based setting (ie, requiring machine equipment), were excluded. While data in symptomatic patients with hip or lower limb pathology were excluded, data of healthy subjects that were collected as part of comparative studies was included. To better standardize and compare outcomes across research, studies that did not normalize EMG activity to an MVIC undertaken in side lying (ie, standing or supine) were excluded. Finally, conference abstracts, non-peer-reviewed studies, case reports and reviews were also excluded.

## Study Review Process

Two authors (JE and PE) independently reviewed the literature for all titles and abstracts according to the aforementioned criteria. Any disagreements were resolved via a post review discussion, reanalysis of the nominated study and final consensus. All articles considered appropriate could be sourced in full, and were read in their entirety to establish if they met the eligibility criteria.

## Data Extraction

Following consensus on the final selection of studies, one author (JE) extracted the required information from each study including subject characteristics, method of standardizing EMG data, the evaluated limb (dominant, nondominant, left and/or right), the reported protocol of subject evaluation, the method of EMG collection and the exercises evaluated (with %MVIC and SD if reported). Subsequent to this, reported exercises were stratified based on exercise type, including 1) prone and quadruped exercises, 2) bridging exercises (supine, prone, and side lying), 3) specific hip abduction and rotation exercises (standing or side lying), 4) standing weight-bearing (WB) exercises, and 5) functional WB exercises. Based on these subgroupings, we further stratified exercises into low (0% to 20% MVIC), moderate (21% to 40% MVIC), high (41% to 60% MVIC), and very high (> 61% MVIC) loading groups.[19,20] Rather than providing pooled means (SD) of each exercise evaluated across different studies, individual means were presented to permit an appreciation of the range (and variability) of %MVIC reported for the same exercises evaluated, as well as variations of a particular type of exercise.

# Evidence Synthesis

## Study Selection

Figure 1 demonstrates the flowchart of study search and final selection. The initial database search yielded 2074 articles, refined to 435 after exclusion of duplicates. We identified 41 studies worthy of full text review after screening all manuscript titles and abstracts and, after applying the inclusion/exclusion criteria and consultation among the 2 separate reviewers, 20 studies were selected for the final review.

## Characteristics of Included Studies

Characteristics of the 20 included studies are summarized in Table 1. In total, 438 asymptomatic subjects were evaluated within these studies with a varied underlying activity history. Apart from 2 studies that included males

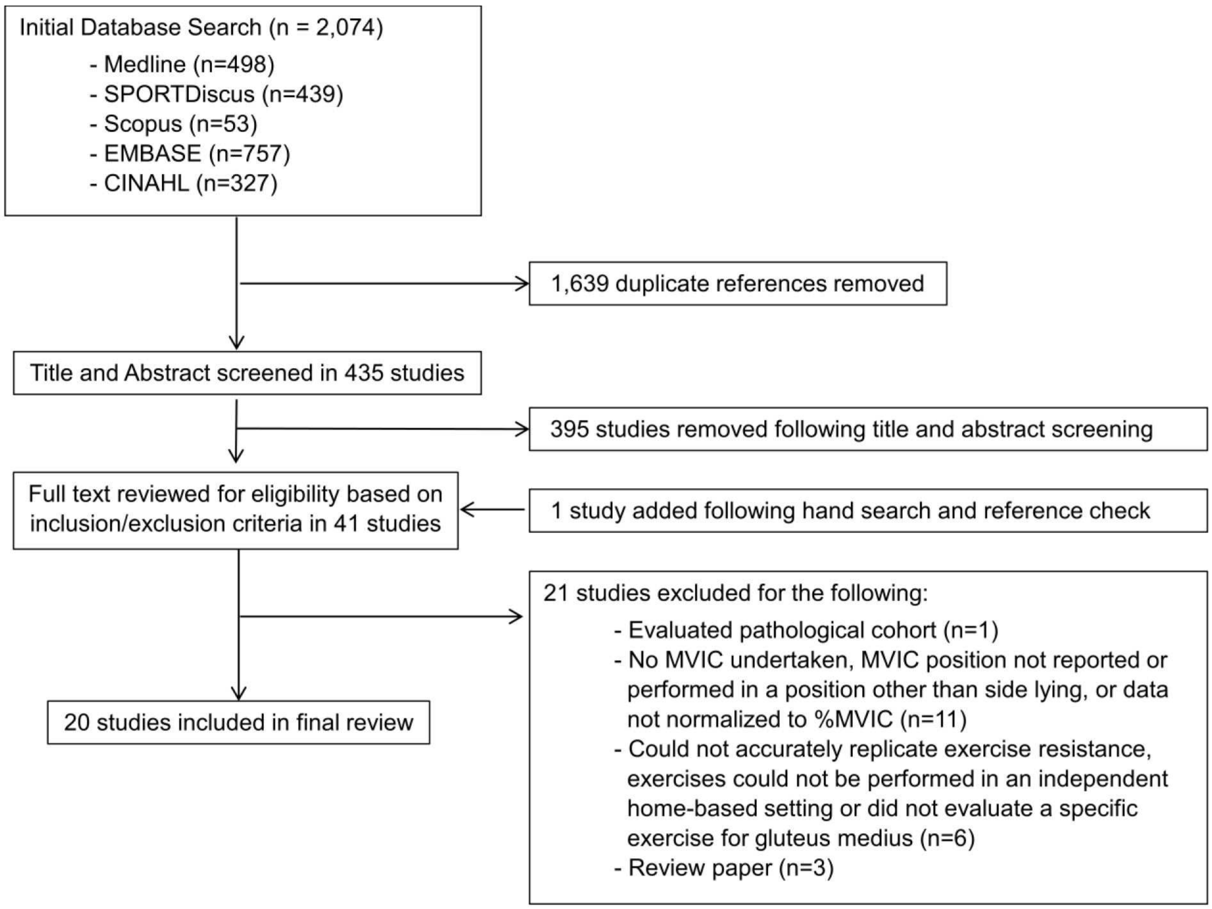


**Figure 1** — Flowchart of study search and selection process.

only,[22,23] 1 study that included females,[13] and 1 study which did not report gender distribution,[11] the remaining 16 studies evaluated exercises in a spread of males and females. The range of subject ages was generally not reported within studies, though the group means of each study ranged from 21.0–31.2 years. While 3 studies specifically reported evaluation of the right limb,[22–24] 2 studies reported evaluation of the left and/or right,[10,18] and in 1 study it was not reported,[6] the remaining 14 studies evaluated the dominant limb (generally the limb used to kick a ball). Of the 20 included studies, 19 employed surface EMG to collect gluteus medius muscle activation, with the final study using fine-wire EMG.[25] A total of 21 studies were excluded following full text review[19,20,26–44] for reasons outlined in Figure 1.

## Summary of Exercises Evaluated

Of the 20 included studies, 33 exercises (not including a range of variations of the same exercise) were evaluated. A summary of investigated exercises according to exercise type and gluteus medius activation (%MVIC) is provided in Table 2.

***Prone and Quadruped Exercises.*** Two exercises were evaluated including prone hip extension with a flexed knee (moderate, 38% MVIC) and 4 variations of hip extension in a quadruped position investigating the WB and non-WB limbs (moderate-high, 22%–47% MVIC) (Table 2).

***Bridging Exercises.*** Six bridging exercises (and variations) were evaluated in a range of supine, prone and side lying positions. These included a bilateral supine bridge (low-moderate, 15%–28% MVIC), a unilateral supine bridge on a stable (moderate-high, 31%–55% MVIC) and unstable (high, 47% MVIC) surface, a bilateral prone bridge (moderate, 27% MVIC), a unilateral prone bridge with added hip extension evaluating the WB (very high, 103% MVIC) and non-WB (very high, 75% MVIC) limbs, a side bridge (very high, 74% MVIC), and a unilateral side bridge with added hip abduction evaluating the WB (very high, 103% MVIC) and non-WB (very high, 89% MVIC) limbs.

***Hip Abduction and Rotation Exercises.*** Three specific hip abduction/rotation exercises (and variations) were evaluated in a range of standing and side lying positions.

**Table 1 Characteristics of Included Studies**

| Study | Sample | Method of standardizing EMG data | Evaluated limb | Protocol sequence | Method of gluteus medius EMG collection | Exercises evaluated and results (% maximal volitional isometric contraction ± SD) |
|---|---|---|---|---|---|---|
| Ayotte et al (2007) | 23 healthy, asymptomatic subjects (16 male, 7 female; mean age 31.2 ± 5.8 years; mean body weight 77.0 ± 13.9 kg; mean height 173.1 ± 10.1 cm). | MVIC testing in side lying, in a position of 0° hip abduction and neutral hip flexion/ extension. | Dominant limb (leg used to kick a ball). | Submaximal cycle warm-up (10 mins), electrode placement, practice and familiarization, MVIC testing and exercise testing. | Surface EMG (33% of the distance from the greater trochanter to the iliac crest. | Undertaken in a randomized order: concentric phase of a: 1) unilateral wall squat (52 ± 22%), 2) unilateral mini-squat (36 ± 17%), 3) forward step-up (44 ± 17%), 4) lateral step-up (38 ± 18%) and 5) retro step-up (37 ± 18%). |
| Barton et al (2014) | 19 healthy asymptomatic subjects (11 male, 8 female) with a mean age of 28.4 ± 2.7 years; mean body weight 67.8 ± 10.4 kg; mean height 172.4 ± 5.8 cm). | MVIC testing in side lying, with the hip in neutral and 10° abduction. | Dominant limb. | Electrode placement, warm-up (5 min brisk walk), MVIC testing, exercise instruction and practice, exercise testing. | Surface EMG, 50% (half of the distance) between the iliac crest and the greater trochanter. | Undertaken in a randomized order: 1) bilateral squat (sliding against the wall) to 90° of knee flexion (9 ± 5), 2) bilateral squat (rolling on a 55 cm swiss ball against the wall) to 90° of knee flexion (10 ± 7), 3) unilateral squat with the non-WB limb receiving support from a wall in a hip abducted and flexed position (42 ± 12), 4) unilateral squat with the non-WB limb receiving support from a 55 cm swiss ball against a wall in a hip abducted and flexed position (46 ± 15). The EMG during the isometric phase of these exercises was evaluated. |
| Berry et al (2015) | 24 healthy asymptomatic subjects (12 male, 12 female) with a mean age of 22.9 ± 2.9 years; mean body weight 68.6 ± 12.9 kg; mean height 171.1 ± 10.5 cm). | MVIC testing in side lying, with the hip in neutral and 10° abduction. | Left and right. | Electrode placement, MVIC testing, exercise instruction and practice, exercise testing. | Surface EMG, over the posterior portion of the gluteus medius muscle belly. | Undertaken in a randomized order: 1) the non-WB (moving) limb during a resisted theraband side step in an upright trunk position (18.7 ± 8.0), 2) the WB (stance) limb during a resisted theraband side step in an upright trunk position (22.9 ± 9.5), 3) the non-WB (moving) limb during a resisted theraband side step in a self-selected squat position (23.3 ± 11.2), 4) the WB (stance) limb during a resisted theraband side step in a self-selected squat position (35.7 ± 13.8). All exercises begun with the feet initially 30 cm apart and with a theraband wrapped around the ankles on a gentle stretch (about 110% of unstretched length), and then proceeding to step out a further 30 cm with each side step. |

*(continued)*

**Table 1 *(continued)***

| Study | Sample | Method of standardizing EMG data | Evaluated limb | Protocol sequence | Method of gluteus medius EMG collection | Exercises evaluated and results (% maximal volitional isometric contraction ± SD) |
|---|---|---|---|---|---|---|
| Bolgla et al (2005) | 16 healthy, asymptomatic subjects (8 male, 8 female; mean age 27 ± 5 years; mean body weight 76 ± 15 kg; mean height 1.7 ± 0.2 m). | MVIC testing in side lying, in a position of 25° hip abduction. | Right limb. | Submaximal cycle warm-up (5 mins), gentle lower limb stretching exercises, exercise familiarization, 10 minute rest period, electrode placement, MVIC testing, exercise testing, reevaluation of MVIC. | Surface EMG, one-third the distance between the iliac crest and greater trochanter over the right gluteus medius muscle belly. | Undertaken in a randomized order (with a cuff mass equal to 3% of the individual's body mass was applied to the right test limb for all NWB exercises): 1) standing hip abduction with neutral hip position and test limb is WB limb (42 ± 27), 2) standing hip abduction with neutral hip position and test limb is NWB limb (33 ± 23), 3) standing hip abduction with flexed hip position and test limb is WB limb (46 ± 34), 4) standing hip abduction with flexed hip position and test limb is NWB limb (28 ± 21), 5) side lying hip abduction (42 ± 23) and 6) pelvic drop (57 ± 32). |
| Boren et al (2011) | 26 healthy participants (over the age of 21 years) who were able to exercise for approximately 1 hour | MVIC testing in side lying. | Dominant limb (leg used to kick a ball). | Electrode placement, submaximal cycle warm-up (5 mins), video-based exercise familiarization, MVIC testing, exercise testing, reevaluation of MVIC. | Surface EMG over gluteus medius muscle. | Undertaken in a randomized order (and in bare feet): 1) pelvic drop (58.4), 2) single limb deadlift (56.1), 3) forward step-up (54.6), 4) lateral step-up (59.9), 5) single limb squat (82.3), 6) single limb 'skater' squat (59.8), 7) single limb bridge on a stable surface (55.0), 8) single limb bridge on an unstable surface (47.3), 9) front plank with test limb in bent knee hip extension (75.1), 10) side plank with added hip abduction with test limb as WB limb (103.1), 11) side plank with added hip abduction and test limb is NWB limb (88.8), 12) side lying hip abduction (62.9), 13) standing hip circumduction while standing on the test limb and circumducting the other limb on a stable surface (57.4), 14) standing hip circumduction while standing on the test limb and circumducting the other limb on an unstable surface (37.9), 15) quadruped with added bent knee hip extension and test limb is NWB limb (46.7), 16) quadruped with added bent knee hip extension and test limb is WB limb (22.0), 17) standing maximal gluteal squeeze (43.7), 18) standing hip flexion and extension with test limb as WB limb (57.3) and 4 clamshell variations with the test limb on top, being the 19) standard clamshell in a position of 45° hip flexion (47.2), 20) clamshell in a position of 45° hip flexion in which knees remain together and foot of the test limb lifts via internal hip rotation (62.5), 21) clamshell in a position of 45° hip flexion and held in an abducted position, in which knees remain together and foot of the test limb lifts via internal hip rotation (67.6) and 22) clamshell in a neutral hip in which knees remain together and foot of the test limb lifts via internal hip rotation (76.9). |

*(continued)*

**Table 1** ***(continued)***

| Study | Sample | Method of standardizing EMG data | Evaluated limb | Protocol sequence | Method of gluteus medius EMG collection | Exercises evaluated and results (% maximal volitional isometric contraction ± SD) |
|---|---|---|---|---|---|---|
| Cambridge et al (2012) | 9 healthy, asymptomatic males (mean age 22.6 ± 2.2 years; mean body weight 85.8 ± 15.4 kg; mean height 181.9 ± 9.2 cm). | MVIC testing in side lying. | Right limb. | Electrode placement, MVIC testing, exercise familiarization, exercise testing. | Surface EMG, exact location over gluteus medius muscle not specified. | Undertaken in a randomized order: 1) a 'Sumo Walk', or front plane upright walk in a semisquat position, undertaken with a theraband around the feet (35-40), ankles (25-30) or knees (20-25), and 2) a 'Monster Walk', or sagittal plane upright side-step in a semisquat position, undertaken with a theraband around the feet (25-30), ankles (20-25) or knees (15-20). |
| Cynn et al (2006) | 18 healthy asymptomatic subjects (9 male, 9 female) with a mean age of 23.5 ± 3.5 years; mean body weight 59.3 ± 5.1 kg; mean height 167.7 ± 4.3 cm). | MVIC testing in side lying, with the hip in slight extension and lateral rotation, and in a position of 50% of subject's hip abduction range of motion. | Dominant limb. | Electrode placement, MVIC testing, exercise instruction and practice, exercise testing. | Surface EMG, over the proximal one-third of the distance between the iliac crest and greater trochanter. | Undertaken in a randomized order: 1) side lying hip abduction to 35° (25.03 ± 10.25), 2) side lying hip abduction to 35° in a lumbar stabilized position, using a pressure biofeedback unit (46.06 ± 21.20). The lumbar stabilized exercise was omitted from this review due to the difficulty in undertaking this independently. |
| Distefano et al (2009) | 21 healthy, asymptomatic, recreationally active subjects, participating in physical activity for at least 6 mins, 3 days per week (9 male, 12 female; mean age 22 ± 3 years; mean body weight 70.4 ± 15.3 kg; mean height 171 ± 11 cm). | MVIC testing in side lying, in a position of 25° hip abduction. | Dominant limb (leg used to kick a ball). | Submaximal jogging warm-up (5 mins), exercise familiarization and practice, electrode placement, exercise testing, 5 mins rest, MVIC testing. | Surface EMG (33% of the distance from the greater trochanter to the iliac crest. | Undertaken in a randomized order with 2 mins rest time between exercises: 1) clamshell with 30° hip flexion (40 ± 38), 2) clamshell with 60° hip flexion (38 ± 29), 3) side lying hip abduction (81 ± 42), 4) single limb squat (64 ± 24), 5) single limb deadlift (58 ± 25), 6) lateral band walk (61 ± 34), 7) forward lunge to 90° of hip and knee flexion (42 ± 21), 8) transverse lunge to 90° of hip and knee flexion (48 ± 21), 9) side lunge to 90° of hip and knee flexion (39 ± 19), 10) forward hop of half body height length, jumping off the nondominant limb and landing on the dominant/test limb (45 ± 21), 11) transverse hop of half body height length, jumping off the nondominant limb and landing on the dominant/test limb (48 ± 25), 12) side hop of half body height length, jumping off the nondominant limb and landing on the dominant/test limb (57 ± 35). |

*(continued)*

**Table 1 *(continued)***

| Study | Sample | Method of standardizing EMG data | Evaluated limb | Protocol sequence | Method of gluteus medius EMG collection | Exercises evaluated and results (% maximal volitional isometric contraction ± SD) |
|---|---|---|---|---|---|---|
| Ekstrom et al (2007) | 30 healthy, asymptomatic subjects (19 male, 11 female; mean age 27 ± 8 years; mean body weight 74 ± 11 kg; mean height 176 ± 8 cm). | MVIC testing in side lying, with the hip in neutral rotation and slight extension. | Left or right, not controlled. | Exercise familiarization and practice, electrode placement, MVIC testing, exercise testing. | Surface EMG, positioned anterosuperior to the gluteus maximus muscle and just inferior to the iliac crest on the lateral side of the pelvis. | Undertaken in a randomized order: 1) side lying hip abduction (39 ± 17), 2) bilateral supine bridge (28 ± 17), 3) unilateral supine bridge (47 ± 24), 4) side bridge with test limb in WB (74 ± 30), 5) prone bridge (27 ± 11), 6) quadruped with straight knee hip extension and contralateral arm lift with test limb in NWB (42 ± 17), 7) lateral step-up onto an 8-inch platform (43 ± 18), 8) forward lunge (29 ± 12). A Dynamic Edge (The Skier's Edge Company, Park City, UT) platform (33 ± 16) was evaluated though omitted from this review due to the requirement of the device. |
| Krause et al (2009) | 14 healthy asymptomatic females (mean age 23.6 ± 1.7 years; mean body weight 65.0 ± 9.2 kg; mean height 169.3 ± 9.5 cm) and 6 males (mean age 26.3 ± 2.5 years; mean body weight 85.0 ± 10.1 kg; mean height 172.2 ± 12.9 cm) | MVIC testing in side lying, with the hip in slight extension and 30° abduction. | Dominant limb (leg used to kick a ball). | Electrode placement, walking warm-up (5 mins), MVIC testing, exercise familiarization, practice and testing. | Surface EMG, 50% (half of the distance) between the iliac crest and the greater trochanter. | Undertaken in a randomized order: 1) double limb stance (≈5), 2) single limb stance (≈20), 3) single limb stance while standing on a foam (unstable) surface (≈25), 4) single limb squat to 45° of knee flexion (≈50), 5) single limb squat to 45° of knee flexion while standing on a foam (unstable) surface (≈60). |
| Lee et al (2013) | 20 healthy asymptomatic subjects (mean age 22.3 ± 1.9 years; mean body weight 65.5 ± 12.4 kg; mean height 168.7 ± 7.2 cm). | MVIC testing in side lying, with the hip in slight extension and lateral rotation, and in a position of 50% of subject's hip abduction range of motion. | Dominant limb (leg used to kick a ball). | Submaximal jogging warm-up (5 mins), session familiarization, electrode placement, MVIC testing, exercise testing. | Surface EMG, directly superior to the greater trochanter, one-third of the distance between the iliac crest and greater trochanter. | Undertaken in a randomized order: 1) side lying hip abduction with neutral hip position (34.2 ± 11.8), 2) side lying hip abduction with maximal medial (internal) hip rotation (45.3 ± 20.5), 3) side lying hip abduction with maximal lateral (external) hip rotation (35.3 ± 12.5). |

*(continued)*

**Table 1 *(continued)***

| Study | Sample | Method of standardizing EMG data | Evaluated limb | Protocol sequence | Method of gluteus medius EMG collection | Exercises evaluated and results (% maximal volitional isometric contraction ± SD) |
|---|---|---|---|---|---|---|
| Lee et al (2014) | 19 healthy subjects (8 male, 11 female) with a mean age 21.00 ± 1.73 years; mean body weight 59.79 ± 9.61 kg; mean height 166.00 ± 0.7 cm); mean body mass index 21.54 ± 2.56 kg/m$^2$. All patients were determined to have weak gluteus medius strength via manual muscle testing. | MVIC testing in side lying, with the hip in slight extension and lateral rotation, and in a position of 50% of subject's hip abduction range of motion. | Dominant limb (leg used to kick a ball). | Submaximal jogging warm-up (5 mins), session familiarization, electrode placement, MVIC testing, exercise testing. | Surface EMG, directly superior to the greater trochanter, one-third of the distance between the iliac crest and greater trochanter. | Undertaken in a randomized order: 1) side lying hip abduction with neutral hip position (45.22), 2) side lying hip abduction with medial (internal) hip rotation equal to 50% of the subject's maximal medial hip rotation range of motion (61.34), 3) side lying hip abduction with lateral (external) hip rotation equal to 50% of the subject's maximal lateral hip rotation range of motion (48.96). |
| Lubahn et al (2011) | 18 healthy asymptomatic females (mean age 22.3 ± 2.3 years; mean body weight 61.1 ± 7.1 kg; mean height 166.8 ± 9.2 cm). | MVIC testing in side lying with the hip in a neutral position. | Dominant limb (leg used to kick a ball). | Electrode placement, MVIC testing, exercise familiarization and testing. | Surface EMG, on the proximal one-third of the distance between the iliac crest and greater trochanter. | Undertaken in a randomized order: 1) double leg squat (17.6 ± 10.4), 2) double leg squat with additional lateral resistance provided by a green theraband (in a 30cm loop) around the subject's knees (18.7 ± 11.1), 3) single leg squat (47.5 ± 13.2), 4) front step up (43.5 ± 14.4). In addition, a single leg squat (40.0 ± 13.8) and a front step up (43.8 ± 20.1) with added resistance provided by a cable column pulling horizontally in a medial direction were evaluated, though omitted from this review given the requirement of a cable machine. |
| MacAskill et al (2014) | 20 healthy asymptomatic females (mean age 21.7 ± 1.6 years; mean body weight 58.1 ± 6.2 kg; mean height 163.2 ± 6.7 cm) and 14 males (mean age 21.2 ± 1.8 years; mean body weight 77.1 ± 8.9 kg; mean height 177.8 ± 15.3 cm), with no recent history (6 months) of lower limb resistance training. | MVIC testing in side lying, with the hip in a position of 50% of subject's hip abduction range of motion. | Dominant limb (leg used to kick a ball). | Exercise familiarization (2 weeks before data collection), 10 repetition maximum testing (1 week before data collection), electrode placement, MVIC testing, exercise testing. | Surface EMG, 2–3 cm distal to the midpoint of the iliac crest. | Undertaken in a randomized order: 1) forward step up onto a 6 inch box (63 ± 18), 2) lateral step up onto a 6 inch box (61 ± 20), 3) 10 RM side lying hip abduction (100 ± 17), 4) 10 RM bent knee (90°) prone hip extension (38 ± 22). The 10 RM for the NWB exercises was provided by external load applied to the lower extremity. |

*(continued)*

Table 1 *(continued)*

| Study | Sample | Method of standardizing EMG data | Evaluated limb | Protocol sequence | Method of gluteus medius EMG collection | Exercises evaluated and results (% maximal volitional isometric contraction ± SD) |
|---|---|---|---|---|---|---|
| McBeth et al (2012) | 20 healthy asymptomatic distance runners (average 40km per week in the 6 weeks before testing), including 11 females (mean age 26.1 ± 5.2 years; mean body weight 61.3 ± 6.6 kg; mean height 1.68 ± 0.03 m; mean body mass index 21.7 ± 1.5 kg/m$^2$) and 9 males (mean age 26.6 ± 6.5 years; mean body weight 69.3 ± 7.1 kg; mean height 1.75 ± 0.08 m; mean body mass index 22.6 ± 1.2 kg/m$^2$). | MVIC testing in side lying, with the hip in slight extension and external rotation, and in a position of 35° of hip abduction. | Dominant limb (leg used to kick a ball). | Exercise instruction and practice, warm-up (5 min moderate jog on a treadmill), electrode placement, MVIC testing, exercise testing. | Surface EMG, directly superior to the greater trochanter, one-third of the distance between the iliac crest and greater trochanter. | Undertaken in a randomized order: 1) side lying hip abduction to 35° with a neutral hip position and a cuff weight of 5% BW placed around the subject's ankle (79.1 ± 29.9), 2) side lying hip abduction to 35° with maximal external hip rotation and a cuff weight of 5% BW placed around the subject's ankle (53.0 ± 28.4), 3) clam shell exercise to 25°, in a position of 45° hip flexion and 90° knee flexion, with a cuff weight of 5% BW placed proximal to the subject's knee (32.6 ± 16.9). |
| Nakagawa et al (2012) | 20 healthy asymptomatic females (mean age 21.8 ± 2.6 years; mean body weight 59.4 ± 7.3 kg; mean height 1.63 ± 0.73 m) and 20 males (mean age 23.5 ± 3.8 years; mean body weight 74.6 ± 9.1 kg; mean height 1.76 ± 0.61 m). A test group with patellofemoral pain syndrome was also evaluated, though excluded as part of this review. | MVIC testing in side lying, in a position of 0° (neutral) hip abduction. | Not reported, though was matched to a comparative cohort of patients with patellofemoral pain syndrome. | Warm-up (5 min walk on a treadmill at 1.66 m/s), electrode placement, MVIC testing, exercise instruction and practice, exercise testing. | Surface EMG, 50% (half of the distance) between the iliac crest and the greater trochanter. | Single leg squat (males 17.0 ± 5.4; females 29.4 ± 5.5). |

*(continued)*

**Table 1 *(continued)***

| Study | Sample | Method of standardizing EMG data | Evaluated limb | Protocol sequence | Method of gluteus medius EMG collection | Exercises evaluated and results (% maximal volitional isometric contraction ± SD) |
|---|---|---|---|---|---|---|
| Selkowitz et al (2013) | 20 healthy asymptomatic subjects (10 male, 10 female) with a mean age of 27.9 ± 6.2 years. | MVIC testing in side lying, with the hip in 0° of flexion and in a position of 30° of hip abduction. | Dominant limb (leg used to kick a ball). | Electrode placement, MVIC testing, exercise familiarization, practice and testing. | Fine-wire EMG, inserted 2.5 cm distal to the midpoint of the iliac crest. | Undertaken in a randomized order: 1) side lying hip abduction (43.5 ± 14.7), 2) clam shell with a theraband around the thighs (26.7 ± 18.0), 3) bilateral supine bridge (15.0 ± 10.5), 4) unilateral supine bridge (30.9 ± 20.7), 5) hip extension in quadruped on elbows with knee extending (27.3 ± 14.9), 6) hip extension in quadruped on elbows with knee flexed (30.9 ± 15.2), 7) forward lunge (19.3 ± 12.9), 8) squat (9.7 ± 7.3), 9) sidestep with elastic resistance around the thighs in a squatted position (30.2 ± 15.7), 10) hip hike (37.7 ± 15.1), 11) forward step-up (29.5 ± 14.9). |
| Sidorkewicz et al (2014) | 13 healthy asymptomatic males (mean age of 24.8 ± 4.2 years; mean body weight 75.9 ± 9.8 kg; mean height 179.7 ± 5.4 cm). | MVIC testing in side lying. | Right limb. | Exercise familiarization and practice, electrode placement, MVIC testing, exercise testing. | Surface EMG approximately over the middle fibers, 50% (half of the distance) between the iliac crest and the greater trochanter. | Undertaken in a randomized order: 1) side lying hip abduction in a neutral hip position (36.70 ± 14.55), 2) side lying hip abduction in an externally rotated hip position (36.50 ± 16.46), 3) side lying hip abduction in an internally rotated hip position (48.67 ± 20.21), 4) clam shell in a position of 30° of hip flexion (26.80 ± 24.08), 5) clam shell in a position of 45° of hip flexion (35.55 ± 34.25), 6) clam shell in a position of 60° of hip flexion (36.49 ± 33.06). |
| Willcox et al (2013) | 27 (17 with acceptable EMG data that could be included including 10 males and 7 females) healthy asymptomatic subjects (mean age 22.3 ± 1.9 years; mean body weight 65.5 ± 12.4 kg; mean height 168.7 ± 7.2 cm), who participated in moderate-intensity exercise for at least 60 minutes, 3 days per week. | MVIC testing in side lying, with the knee in 90° of flexion. | Dominant limb (leg used to kick a ball). | Electrode placement, MVIC testing, exercise testing. | Surface EMG, one-third of the distance from the greater trochanter to the iliac crest. | Undertaken in a randomized order: 1) clam exercise with the knees flexed to 90° and hip in a neutral position with the hips flexed to 0° (15-20), 30° (20-25) and 60° (20-25), 2) clam exercise with the knees flexed to 90° and hip in a 35° reclined position with the hips flexed to 0° (10-15), 30° (10-15) and 60° (15-20). |

*(continued)*

**Table 1 *(continued)***

| Study | Sample | Method of standardizing EMG data | Evaluated limb | Protocol sequence | Method of gluteus medius EMG collection | Exercises evaluated and results (% maximal volitional isometric contraction ± SD) |
|---|---|---|---|---|---|---|
| Youdas et al (2013) | 21 healthy asymptomatic subjects including 10 males (mean age of 25.0 ± 3.1 years; mean body weight 82.2 ± 7.9 kg; mean height 1.8 ± 0.1 m; mean body mass index 25.0 ± 2.6 kg/m$^2$) and 11 females (mean age of 24.5 ± 1.4 years; mean body weight 69.1 ± 4.9 kg; mean height 1.7 ± 0.1 m; mean body mass index 23.8 ± 2.4 kg/m$^2$). | MVIC testing in side lying, with the hip in a position of 30° of hip abduction. | Dominant limb (leg used to kick a ball). | Electrode placement, MVIC testing, exercise familiarization and practice, exercise testing. | Surface EMG, over the muscle belly (middle fibers). | Undertaken in a randomized order: 1) lateral band walk with neutral hip rotation (moving limb 32.8 ± 21.9; stance limb 49.9 ± 21.9), 2) internal hip rotation (moving limb 43.8 ± 27; stance limb 57.8 ± 24.3), 3) external hip rotation (moving limb 27.3 ± 18.1; stance limb 47.6 ± 21.5). For all exercises, the band was placed around the subject's ankles and the subject was in a position of 20°–30° hip and knee flexion. The exercise was initiated in a stance width equal to individual shoulder width and stepping out with the dominant limb a distance of 160% shoulder width. |

Abbreviations: WB, weight bearing; NWB, non weight bearing; MVIC, maximum volitional isometric contraction; EMG, electromyography.

**Table 2 The Percentage (±SD) of Electromyographic Maximum Volitional Isometric Contraction (%MVIC) Reported for Gluteus Medius During a Range of Therapeutic Rehabilitation Exercises**

| Exercise | Low (0%–20% MVIC) | Moderate (21%–40% MVIC) | High (41%–60% MVIC) | Very high (≥61% MVIC) |
|---|---|---|---|---|
| Prone & quadruped exercises | | | | |
| Prone hip extension with flexed knee (90°) | | 38C (±22)[58] | | |
| Quadruped with straight leg hip extension (test limb NWB) | | 27 (±15)[25] | | |
| Quadruped with bent knee hip extension (test limb NWB) | | 31 (±15)[25] | 47[11] | |
| Quadruped with bent knee hip extension (test limb WB) | | 22[11] | | |
| Quadruped with straight leg hip extension and contralateral arm lift (test limb NWB) | | | 42 (±17)[18] | |
| Bridging exercises (supine, prone, & side lying) | | | | |
| Bilateral supine bridge | 15 (±11)[25] | 28 (±17)[18] | | |
| Unilateral supine bridge (on stable surface) | | 31 (±21)[25] | 47 (±24)[18]; 55[11] | |
| Unilateral supine bridge (on unstable surface) | | | 47[11] | |
| Side bridge (test limb WB) | | | | 74 (±30)[18] |
| Double limb prone bridge | | 27 (±11)[18] | | |
| Single limb prone bridge with added bent knee hip extension (test limb NWB) | | | | 75[11] |
| Single limb prone bridge with added bent knee hip extension (test limb WB) | | | | 103[11] |
| Single limb side bridge with additional hip abduction (test leg NWB) | | | | 89[11] |
| Single limb side bridge with additional hip abduction (test leg WB) | | | | 103[11] |
| Specific hip abduction/rotation exercises | | | | |
| Standing hip abduction in neutral hip position (test limb NWB) | | 33 (±23)[24] | | |
| Standing hip abduction in flexed hip position (test limb NWB) | | 28 (±21)[24] | | |
| Standing hip abduction in neutral hip position (test limb WB) | | | 42 (±27)[24] | |
| Standing hip abduction in flexed hip position (test limb WB) | | | 46 (±34)[24] | |
| Side lying hip abduction | | 25 (±10);[53] 34 (±12);[54] 35B (±12.5);[54] 37J (±15);[23] 37 J,B (±17);[23] 39 (±17)[18] | 42 (±23);[24] 44 (±15);[25] 45B,55; 45A (±21);[54] 49;[55] 49J,A (±20);[23] 53B,C (±28)[56] | 61A,55; 63;[11] 79C (±30);[56] 81 (±42);[12] 100C (±17)[58] |
| Clamshell exercise | 10–25[50] | 27C (±18);[25] 27–37J,23; 33C (±17);[56] 38 (±29);[12] 40 (±38);[12] | 47[11] | |
| Clamshell exercise variations | | | | 63–77[11] |
| WB exercises (standing) | | | | |
| Double limb stance | ≈5[49] | | | |
| Single limb stance (stable surface) | ≈20[49] | | | |
| Single limb balance (unstable surface) | | ≈25[49] | | |

*(continued)*

**Table 2 *(continued)***

| Exercise | Low (0%–20% MVIC) | Moderate (21%–40% MVIC) | High (41%–60% MVIC) | Very high (≥61% MVIC) |
|---|---|---|---|---|
| WB exercises (standing) | | | | |
| Pelvic drop (hip hikes) | | 38 (±15)[25] | 57 (±32)[24]; 58[11] | |
| Standing maximal gluteal squeeze | | | 48[11] | |
| Standing hip flexion/extension (test limb WB) | | | 57[11] | |
| Standing hip circumduction on a stable surface (test limb WB) | | | 57[11] | |
| Standing hip circumduction on a unstable surface (test limb WB) | | 38[11] | | |
| WB exercises (functional) | | | | |
| Double limb free standing squat | 10 (±7);[25] 18[I] (±10);[13] 19 [I,C] (±11)[13] | | | |
| Double limb wall (or swiss ball) assisted squat | 9 (±5)[57], 10 (±7)[57] | | | |
| Single limb wall squat (stable surface) | | | 52 (±22)[9] | |
| Single limb free-standing squat (stable surface) | 17[I] (±5)[6] | 29[J] (±6);[6] 36 (±17)[9] | 42[D] (±12);[57] 46[D] (±15);[57] 48[I] (±13);[13] ≈50[49] | 64 (±24)[12]; 82[11] |
| Single limb free-standing squat (unstable surface) | | | ≈60[49] | |
| Single limb 'skater' squat | | | 60[11] | |
| Single limb deadlift | | | 56[11]; 58 (±25)[12] | |
| Lateral plane band walk ('crab walk' or 'monster walk') | 19[E,G] (±8)[10] | 15–25[J,22], 23[F,G] (±10);[10] 23[E,H] (±11);[10] 27[E,B] (±18);[52] 30 (±16);[25] 33[E] (±22);[52] 36[F,H] (±14)[10] | 44[E,A] (±27);[52] 48[F,B] (±22);[52] 50[F] (±22);[52] 58 [F,A] (±24)[52] | 61 (±34)[12] |
| Frontal plane band walk ('sumo walk') | | 20–40[J,22] | | |
| Forward lunge | 19 (±13)[25] | 29 (±12)[18] | 42 (±21)[12] | |
| Sideways lunge | | 39 (±19)[12] | | |
| Transverse lunge | | | 48 (±21)[12] | |
| Front step up | | 30 (±15)[25] | 44 (±17)[9]; 55[11] | 63 (±18)[58] |
| Front step up and over | | | 44[L] (±14)[13] | |
| Retro step up | | 37 (±18)[9] | | |
| Lateral step up | | 38 (±18)[9] | 60[11]; 43 (±17)[18] | 61 (±20)[58] |
| Forward hop (land on test limb) | | | 45 (±21)[12] | |
| Sideways hop (land on test limb) | | | 57 (±35)[12] | |
| Transverse hop (land on test limb) | | | 48 (±25)[12] | |

Abbreviations: WB, weight bearing; NWB, non weight bearing.

[A] performed in medial hip rotation; [B] performed in lateral hip rotation; [C] performed with a defined additional weight (weighted cuff or theraband); [D] performed with the contralateral limb balanced against a wall or swiss ball; [E] non-WB (moving) limb was evaluated; [F] WB (stance) limb was evaluated; [G] upright trunk position was evaluated; [H] self-selected squat position was evaluated; [I] evaluated in females specifically; [J] = evaluated in males specifically.

These included: standing hip abduction in a neutral or flexed hip position, evaluating both the WB (neutral hip abduction high, 42% MVIC; flexed hip abduction high, 46% MVIC) and non-WB (neutral hip abduction moderate, 33% MVIC; flexed hip abduction moderate, 28% MVIC) limbs, side lying hip abduction performed in a position of neutral (moderate-very high, 25%–100% MVIC), medial (high-very high, 45%–61% MVIC) or lateral (moderate-high, 35%–53% MVIC) hip rotation, and the clam (low-high, 10%–47% MVIC). Four additional variations of the clam exercise were also investigated (very high, 63%–77% MVIC).

***Standing Weight-bearing Exercises.*** Five standing exercises (and variations) were evaluated, including double limb stance (low, 5% MVIC) and double limb stance with a voluntary maximal gluteal squeeze (high, 48% MVIC), single limb stance on a stable (low, 20% MVIC) and unstable (moderate, 25% MVIC) surface, pelvic drops (moderate-high, 38%–58% MVIC), standing hip flexion/extension evaluating the WB limb (high, 57% MVIC), and standing hip circumduction evaluating the WB limb while circumducting around a stable (high, 57% MVIC) or unstable (moderate, 38% MVIC) surface.

***Functional Weight-bearing Exercises.*** A total of 17 functional WB exercises (and variations) were evaluated, including a bilateral free standing squat (low, 10%–19% MVIC); a bilateral wall/ball assisted squat (low, 9%–10% MVIC); a unilateral wall squat (high, 52% MVIC); a unilateral free standing squat on a stable (low-very high, 17%–82% MVIC) and unstable (high, 60% MVIC) surface; a single limb skater squat (high, 60% MVIC); a single limb deadlift (high, 56%–58% MVIC); a lateral plane band walk investigating the WB and non-WB limbs in an upright or flexed trunk position (low-very high, 19%–61% MVIC); a frontal plane band walk (moderate, 20%–40% MVIC); a forward (low-high, 19%–42% MVIC); side (moderate, 39% MVIC) and transverse (high, 48% MVIC) lunge; a forward (moderate-very high, 30%–63% MVIC); retro (moderate, 37% MVIC) and lateral (moderate-very high, 38%–61% MVIC) step up; and a forward (high, 45% MVIC), side (high, 57% MVIC) and transverse (high, 48% MVIC) hop.

## Discussion

This review aimed to provide the clinician with an evidence-based series of exercises, stratified by exercise type and muscular demand (%MVIC), that can be selected in tailoring a graduated gluteus medius loading protocol to a particular patient. It has been previously reported that to achieve strength gains, muscular activation levels > 40% MVIC are required.[9,16,17] Therefore, better knowledge in these higher load exercises will allow the therapist to prescribe the most appropriate activities to load gluteus medius, which may be of benefit toward the later stages of rehabilitation. However, we must also appreciate and be able to intervene with low load exercises when appropriate, such as in patients with significant hip abductor weakness, gluteal pathology and/or following surgical repair. In these patients, improving muscular endurance, preventing physical de-conditioning and/or facilitating motor control and neuromuscular activation[9,18] should be prioritized in the early stages. Exercise prescription must be based on appropriate tissue healing timeframes, patient tolerance and/or progressive overload principles,[45] to avoid aggravating pain with gluteal tendon pathology or jeopardizing the early integrity of a surgical reconstruction. As such and, consistent with previous studies,[42,43] this review classified exercises into low (0% to 20% MVIC), moderate (21% to 40% MVIC), high (41% to 60% MVIC), and very high (> 61% MVIC) loading groups.

This review has shown that gluteus medius EMG muscular activation varied depending on the exercise type, position, complexity and surface stability employed. Exercises evaluated in the prone or quadruped position were moderate loading exercises (22% to 38% MVIC), with the exception of the quadruped bent knee hip extension (47% MVIC) and the quadruped straight knee hip extension with contralateral arm lift (42% MVIC), both evaluating the non-WB limb, which were in the high range. While the aforementioned exercises were lower load activities, clinically these exercises do not place the patient into any known contraindicated or provocative positions in patients such as that following surgical gluteal tendon repair.

As expected, gluteus medius muscle activation during bridging exercises varied based on body position, surface stability and limb support (unilateral or bilateral). Bilateral supine or prone bridging only created a low-moderate load condition (15% to 31% MVIC), whereas the unilateral bridge was moderate-high (31% to 55% MVIC). Not surprisingly, side bridging created some of the highest load exercises across all studies (74% MVIC), with a unilateral side bridge (with the contralateral non-WB leg forced into hip abduction) creating a %MVIC in excess of 100%. Therefore, while supine bridging may advocated in patients during the earlier rehabilitation stages, strength gains are likely to be minimal. However, this graduated prescription of varied bridging activities may serve as a sound progression toward unilateral and side bridge exercises, reserved for later stages of treatment.

The specific hip abduction and rotation activities appeared the most widely studied exercises and, for this reason, some studies also investigated variations of the same exercise to evaluate how changes in body position and hip rotation affected EMG activity. In standing, hip abduction created a moderate load condition (28% to 33% MVIC) for the non-WB limb, irrespective of whether it was undertaken in a neutral or flexed hip position. However, when this exercise was undertaken and evaluated on the WB limb a high load condition was created (42% to 46% MVIC). This highlights the important role gluteus medius plays in WB, and the need to strengthen the hip abductor musculature to stabilize the pelvis and minimize valgus knee alignment during WB.[9]

In comparison with standing, side lying hip abduction must overcome the gravity barrier and was shown to elicit a broad range of moderate, high and very high load conditions (25% to 100% MVIC). Furthermore, it was shown that a medially rotated hip position elicited a higher %MVIC than lateral rotation. When we consider the use of side hip abduction in patients with gluteal pathology, we must consider its potentially provocative nature as well as relative load. For example, the high tensile load could prove detrimental in patients during the early stages after gluteal tendon reconstruction, though given there is often an enthesopathic component to the underlying pathophysiology associated with gluteal tendinopathy,[46] these positions (and isotonic movements) may prove provocative and counterproductive. Furthermore, the clamshell exercise was generally a moderate loading activity for gluteus medius. However, it may still prove painful and/or detrimental in the earlier stages after gluteal tendon reconstruction surgery due to the relative hip flexion, adduction and internal rotation created by the body posture required, all positions which may increase compression over the greater trochanter (and subsequent repair site).[47,48]

When combining all WB exercises evaluated in this review, single limb stance activities as expected increased muscular demand when compared with bilateral stance, though was further increased when WB on an unstable surface.[49] Furthermore, the addition of contralateral limb movement (hip flexion/extension or circumduction) increased demand again. Pelvic drops were investigated in 3 studies and were reported as a moderate-high (38% to 58% MVIC) load WB exercise. Again, while this exercise may be of benefit in retraining gluteus medius in a WB setting, mimicking the specific role of the hip abductors, the relative hip adduction and reduced pelvis-on-femur angle it creates can also be provocative in patients in the earlier stages of rehabilitation.

Interestingly, WB squats (free standing or wall/ball assisted) investigated across multiple studies produced only a low gluteus medius loading condition (10% to 19% MVIC). A unilateral squat (and variations of) was extensively evaluated, though demonstrated high variability across studies being reported as a low, moderate, high and very high exercise. Due to potential reasons discussed later, subtle differences among studies in the way the unilateral squat exercise was prescribed or undertaken may contribute to this variability. The single limb deadlift also proved a high loading exercise (56% to 58% MVIC) given the stabilizing requirement of gluteus medius. However, while exercises such as single limb deadlifts and pelvic drops appear beneficial to improving gluteus medius strength, patients with concomitant low back pain may be intolerant to these exercises.[22] Therefore, an individualized and progressive loading program should be based not just on the anticipated muscular demand, but the patient's individual circumstances.

The lateral plane band walk was also a heavily investigated activity, with high variability that spanned all loading conditions. This variability again appeared to be the result of a number of factors, including hip (medial or lateral rotation) and trunk (upright or flexed) position, as well as band placement (thighs, knees, ankles, or toes), with a medially rotated hip, flexed trunk and more distal band fixation all associated with greater gluteus medius muscular activity. Finally and, as expected, higher loads were observed with the varied array of lunge (19% to 48% MVIC), step (30% to 63% MVIC), and hop (45% to 57% MVIC) exercises.

As reported above, our review has demonstrated high variability in reported %MVIC for some exercises, across different studies. This is despite the apparent similarity in participant groups with respect to age, gender distribution and activity history, as well as the method selected for evaluating an MVIC in side lying for normalization of the selected exercises. Therefore, while it may still prove difficult in directly comparing gluteus medius activation levels between studies owing to numerous factors including EMG collection and processing methods, limb positioning during the MVICs[11,50] and subtle differences in the way a particular exercise was undertaken, the range of loads (%MVIC) exhibited by a particular exercise across different studies can still be used by the therapist in determining whether that exercise may be appropriate for the patient.

Variations in the standard deviation reported for gluteus medius activation within studies reflects patient individual characteristics and, therefore, what may present as a low load exercise for a well conditioned subject may prove more challenging (and higher load) for a more physically deconditioned subject. Homan et al[51] demonstrated that in a bilateral jump landing, individuals with weaker hip abductor and external rotator strength attempted to counter hip muscle weakness via a heightened recruitment of gluteal muscle activity. We cannot appreciate the baseline strength and conditioning, and past and present activity history of subjects evaluated across the reported studies, given it was generally not reported.

Also has previously suggested, subtle variation in exercise test positions and trunk/pelvic postures for similar exercises (such as the clam, side lying hip abduction, unilateral squat and lateral band walks) may contribute to the variability in %MVIC reported for these exercises. The clamshell exercise has been reported in a position of 0°,[50] 30°,[12,50] 45°,[11] and 60°[12,50] of hip flexion, as well as in a neutral or semireclined (35°) position.[50] Willcox et al[50] demonstrated that gluteus medius muscle activity increased with a neutral hip position, compared with a semireclined (35°) position, as well as with increasing hip flexion angle. Therefore, we can again gauge a loading range of a particular exercise, though direct comparison is made more difficult without an identical exercise description or series of photos.

Of the 20 studies included in this review, 19 employed surface EMG electrodes to collect muscular signal. While this remains common for evaluating gluteus medius, there is always the potential risk of muscular crosstalk despite the use of standardized and optimal electrode anatomical placement. Furthermore, variation existed in the reported position of electrode placement, including: over the gluteus medius muscle;[11,22,52] over the posterior portion of the gluteus medius muscle belly;[10] 33% (or one-third) of the distance from the greater trochanter to the iliac crest;[9,12,13,24,50,53–56] 50% of the distance between the iliac crest and the greater trochanter;[6,23,49,57] 2–3 cm distal to the midpoint of the iliac crest;[58] and anterosuperior to the gluteus maximus muscle and just inferior to the iliac crest on the lateral side of the pelvis.[18] This in itself can create some of the variability observed, and O'Sullivan et al[38] has demonstrated significant variation in muscle activation across the 3 different subdivisions of the gluteus medius (anterior, middle, and posterior segments).

While the aim of this review was to provide the clinician with an evidence-based series of exercises to better tailor a graduated gluteus medius loading protocol to a particular patient, there is also a need to consider the activation of other muscles that may be counterproductive to the rehabilitation process. Fujisawa et al[31] demonstrated that gluteus medius muscular activity during isometric hip abduction was not affected by various angles (0°, 20°, 40°, 60°, or 80°) of hip flexion, though gluteus maximus and tensor fasciae latae (TFL) activity significantly increased and decreased, respectively, with increasing hip flexion angle. Similarly, Lee et al[54,55] have demonstrated that during side lying hip abduction, a medially rotated hip position created greater gluteus medius muscle activity and a greater gluteus medius to TFL muscle activity ratio, while significantly greater TFL muscle activity was created with a laterally rotated hip position. Therefore, understanding not just the relative increase in gluteus medius activity, but the increase/decrease in activity of other muscles contributory to the movement (which may be in favor or not of the underlying program goals), as well as the timing and synergy of muscular cocontraction, is also important. As reported by O'Sullivan et al,[38] studies report the amplitude of gluteus medius muscle activation (%MVIC), though the timing of muscular activation has not been investigated and is an area of future research.

To better standardize and compare the outcomes across studies, those that did not normalize the EMG activity of exercises to an MVIC undertaken in side lying were excluded. While evaluating hip abduction strength in side lying has been frequently employed in clinical settings,[59] a supine (that neutralizes the gravitational effect and provides an option of individuals lying on their injured side)[60] and standing (reported to be more functional as the majority of daily living activities involve hip abduction performed in this position)[61] position has also been employed. Therefore, for the purpose of this review we could not accommodate and compare exercise findings from these other studies, though they may still provide valuable insight into other exercises commonly employed in a rehabilitation setting, though not necessarily covered in the current review.

A final limitation of this review, or at least some caution in the interpretation of the results, is that these studies have been undertaken in asymptomatic and often young, healthy and active subjects. Dwyer et al[29] demonstrated greater gluteus medius muscle activation (%MVIC) during WB step exercises in patients with hip osteoarthritis, compared with matched healthy control subjects. Jacobs et al[32] evaluated gluteus medius EMG activity in patients at least 6 weeks after total hip arthroplasty, during WB and non-WB rehabilitation exercises and, while reported EMG as a %MVIC was relatively high, this was not compared with a healthy control cohort. It has been suggested that pathological groups may present with a biased MVIC evaluation due to the potential for pain,[29] therefore limiting the ability to undertake such studies in a pathological cohort. However, the results of this review can still be interpreted in a way that provides useful applicability to the vast array of patients seen in clinical practice.

## Conclusions

This review has outlined a series of exercises commonly employed in a clinical rehabilitation setting, stratified based on exercise type and %MVIC. While muscular activation levels ≥ 40% MVIC may be required for strength gains,[9,16,17] it has been reported that muscle activity < 25% MVIC may be more important in developing muscular endurance.[17] Furthermore, these low level exercises must be considered when attempting to prevent physical de-conditioning and/or facilitate motor control or neuromuscular activation,[9,18] though without jeopardizing the early integrity of a surgical reconstruction. Across different exercises, variation exists based on the exercise type, whether it is a WB or non-WB activity, whether it is a unilateral or bilateral exercise, the underlying surface on which the exercise is performed, and the relative trunk and/or hip position. Variability across studies investigating the same (or similar) exercises also exists, and particular attention must be paid to the specific exercise prescription, as well as the aforementioned factors, in interpreting and then applying the results in a clinical environment.

## Practice Recommendations

- The results of this review provides the clinician with an evidence-based series of exercises that can be used to develop and apply a graduated loading program for gluteus medius, ranging from the early stages of rehabilitation (ie, postsurgery) through to a higher demand setting.
- These findings must still be interpreted carefully given the variability reported across studies investigating the same (or similar) exercises, with particular

reference to relative trunk and/or hip position and the specifics of exercise prescription. Furthermore, in addition to the relative magnitude of gluteus medius loading, the therapist must also accommodate individual patient characteristics and activity history, pathological and pathophysiological considerations, and adjunct conditions.

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